UNIT-1

DC & AC CIRCUITS

DC circuits: Ohm's and Kirchhoff's laws, analysis of series, parallel and series-parallel circuits excited by independent voltage sources for R, L, C parameters, current division, voltage division.

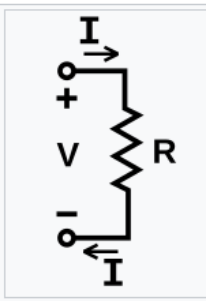
AC circuits: Generation of sinusoidal voltage, frequency of generated voltage, average value, RMS value, form, and peak factors. Real power, reactive power, apparent power, and Power factor.

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1. Define ohm's law and write its limitations?

Statement

Ohm's law states that, at constant temperature, the current flowing through a conductor is directly proportional to the voltage across the element (Resistance)



$$I \propto V \; ; I = \frac{V}{R} \; or \; V = IR$$

Limitations of Ohm's law

- Ohm's law is applicable when the temperature of the conductor is constant. Resistivity changes with temperature.
- The relation between voltage and current depends on the sign of voltage.
- It does not apply to semiconductors, which do not have a direct current-voltage relationship.
- Ohm's law is not applicable for unilateral electrical elements like diodes and transistors as they allow the current to flow through in one direction only.
- For non-linear electrical elements with parameters like capacitance, resistance etc the ratio of voltage and current won't be constant with respect to time making it difficult to use Ohm's law.

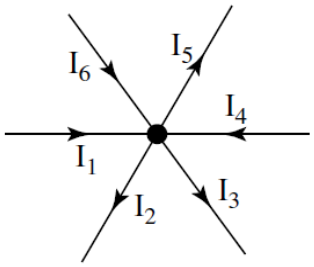
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2. Explain Kirchoff's laws with examples

Kirchoff Current Law:

- The algebraic sum of currents meeting at a junction or a node is equal to zero.
- The sum of current flowing towards a junction or a node is equal to the sum of currents flowing out of the junction.
- The current entering the junction has been taken as positive while the currents leaving the junction have been taken as negative.

Example:



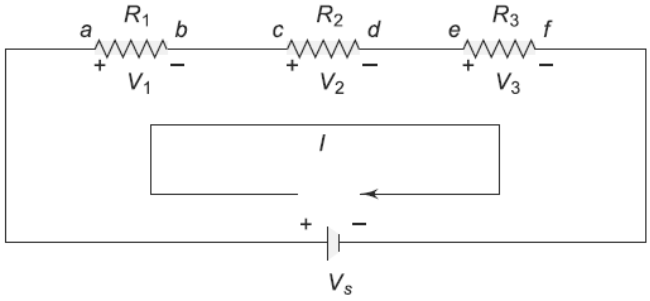
$$I_1 + I_6 + I_4 - I_5 - I_3 - I_2 = 0$$

$$I_1 + I_6 + I_4 = I_5 + I_3 + I_2$$

kirchhoff's voltage law:

- The algebraic sum of voltages in a closed loop is equal to zero.
- The sum of all raising potential(voltage) is equal to the sum of all dropping potential(voltages).

Example:



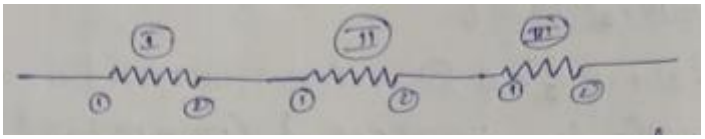
$$V_s = V_1 + V_2 + V_3$$

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3. Series – Parallel combinations of R, L, C

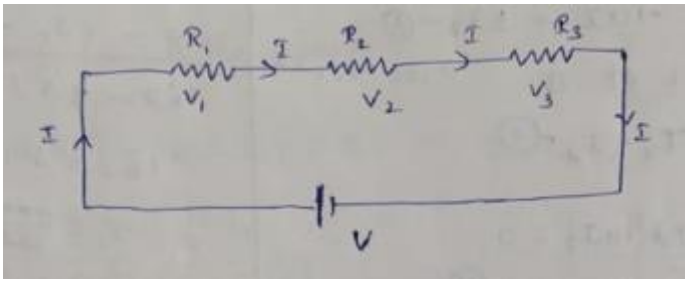
Series connection:

The 2nd terminal of the first element is connected to the 1st terminal of the second element and so on is called series connection.



→ The current at each resistor is always same in series connection.

→ The voltage varies for each resistor

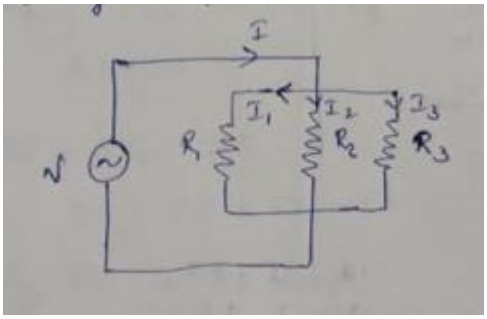


Parallel Connection:

All the first terminals are connected to one end, and the other terminals are connected to the 2nd end is called parallel connection

→ In parallel connection current divides.

→ Voltage across each resistor is same



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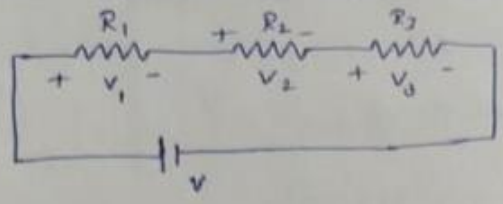
4. Current and voltage relation in resistance, inductance and capacitance

	Voltage	Current
Resistance (R):	V = R I	I = V / R
Inductance (L):	V = L (di/dt)	I = (1/L) ∫ V dt
Capacitance (C):	V = (1/C) ∫ I dt	I = C (dV/dt)

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5. Resistances in Series:

Equivalent resistance in Series Connection



V_1, V_2 & V_3 be the voltage drop across R_1, R_2 & R_3 respectively

According to ohm's law

$V_1 = R_1 I$; $V_2 = R_2 I$; $V_3 = R_3 I$

But we know that by KVL

$V = V_1 + V_2 + V_3$

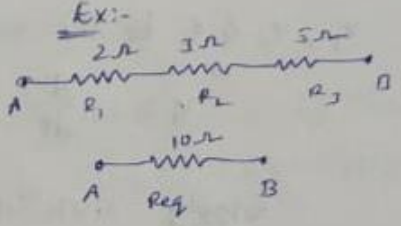
$V = R_{eq} I$

$R_{eq} I = R_1 I + R_2 I + R_3 I$

$I \cdot R_{eq} = I \cdot (R_1 + R_2 + R_3)$

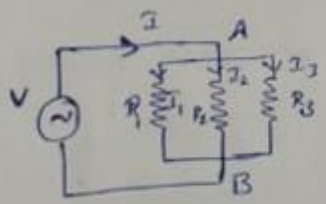
$$\boxed{R_{eq} = R_1 + R_2 + R_3}$$

Ex:-



6. Resistances in Parallel

Equivalent resistance in parallel Connection



I_1, I_2, I_3 are the currents flowing across the resistance R_1, R_2 & R_3 respectively, at A.

apply KCL

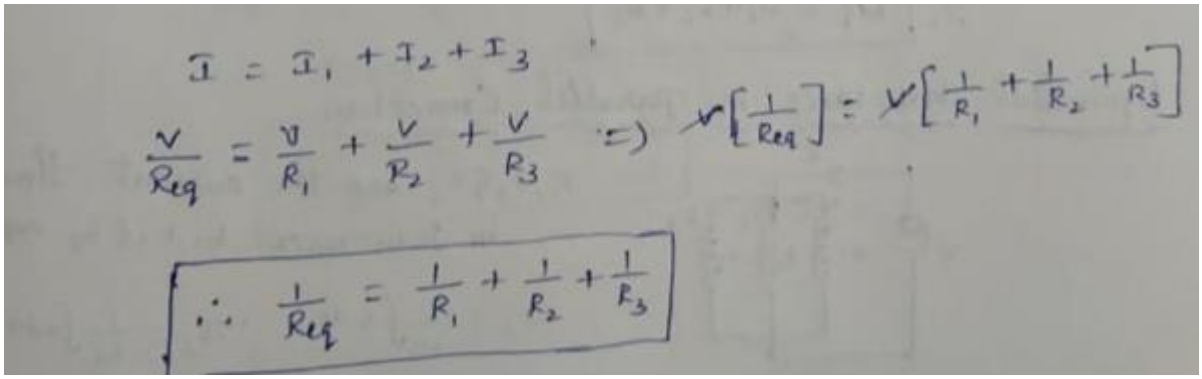
$I = I_1 + I_2 + I_3$ — (1)

Voltage across R_1 is $\Rightarrow V = R_1 I_1 \Rightarrow I_1 = \frac{V}{R_1}$

Voltage across $R_2 \Rightarrow V = R_2 I_2 \Rightarrow I_2 = \frac{V}{R_2}$

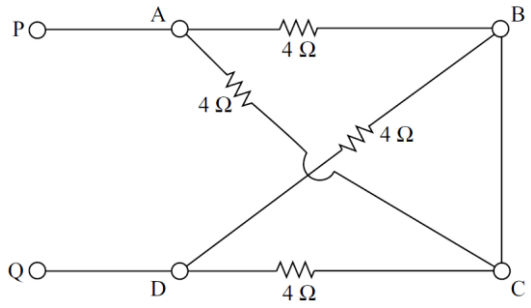
Voltage across $R_3 \Rightarrow V = R_3 I_3 \Rightarrow I_3 = \frac{V}{R_3}$

$V = R_{eq} I \Rightarrow I = \frac{V}{R_{eq}}$



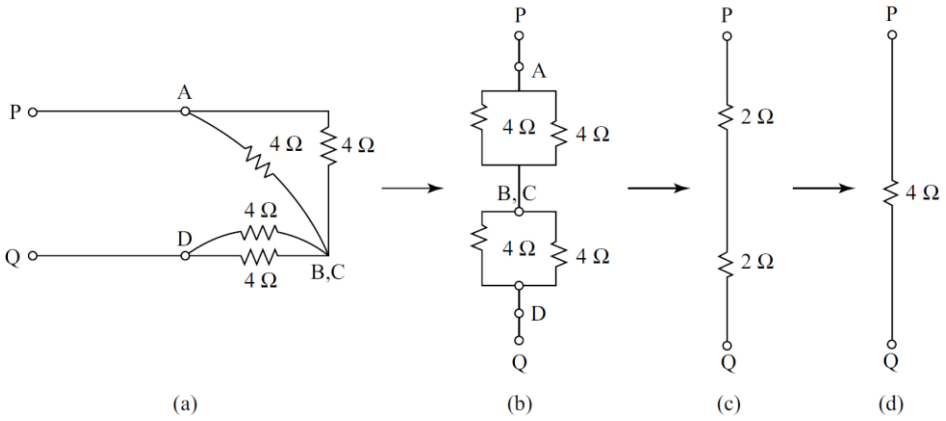
7. Problems:

Example 1. Calculate the resistance between the terminals P and Q of the network.



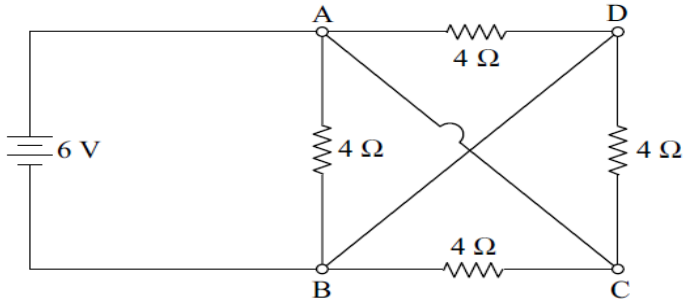
Solution:

There is no resistance in the branch B & C, so Let us bring points B and C together.



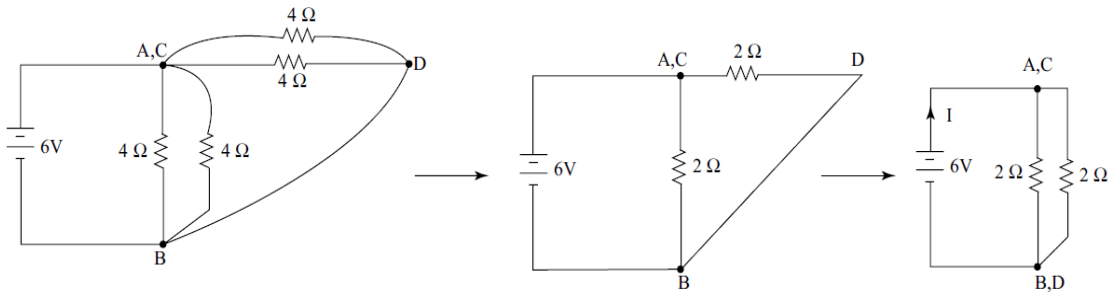
Answer = 4 ohm.

Example 2: Find the Req of the given circuit

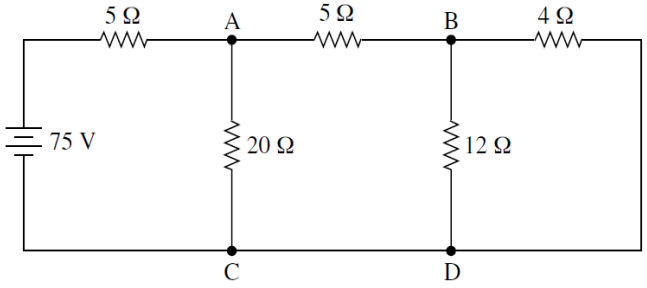


Solution:

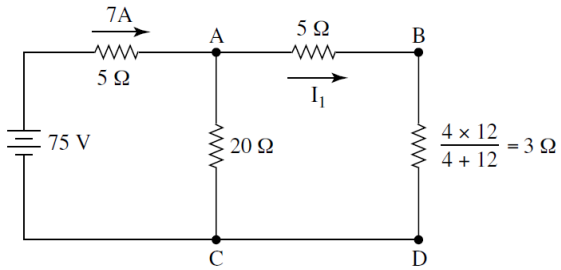
Points A and C are joined together. Similarly points B and D are joined together.



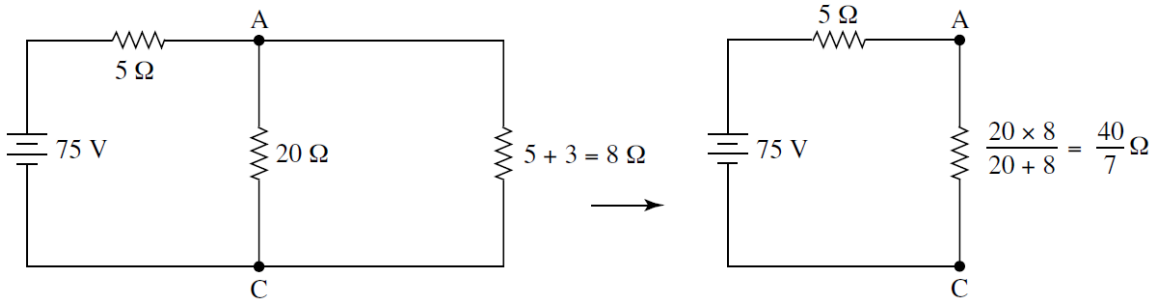
Example 3: Find the R_{eq} of the given circuit



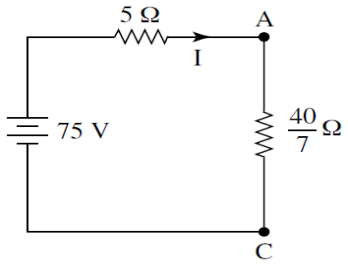
Solution:



(a)



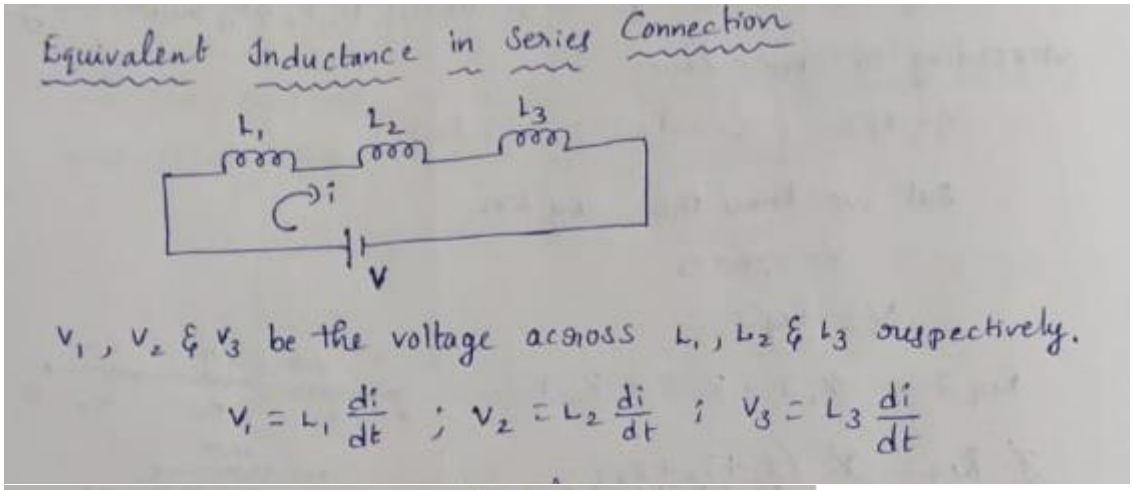
(b) **(c)**

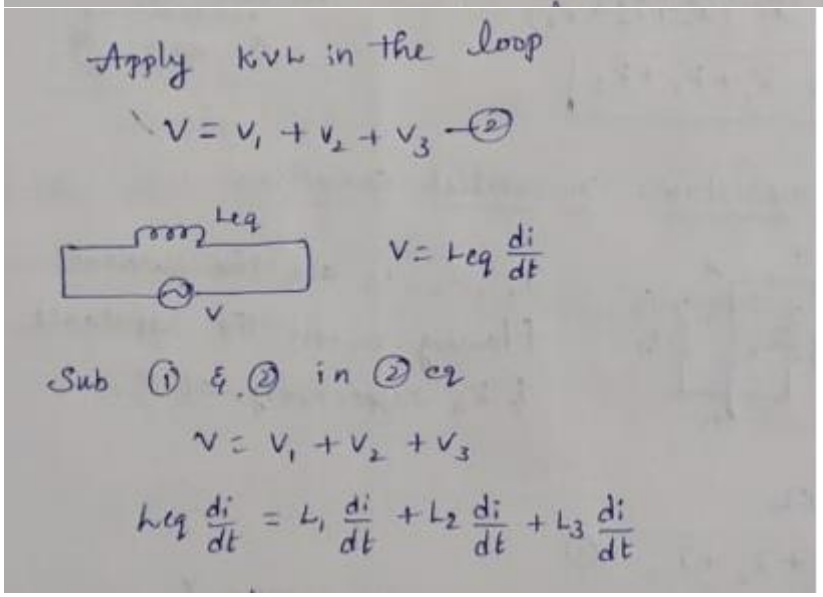


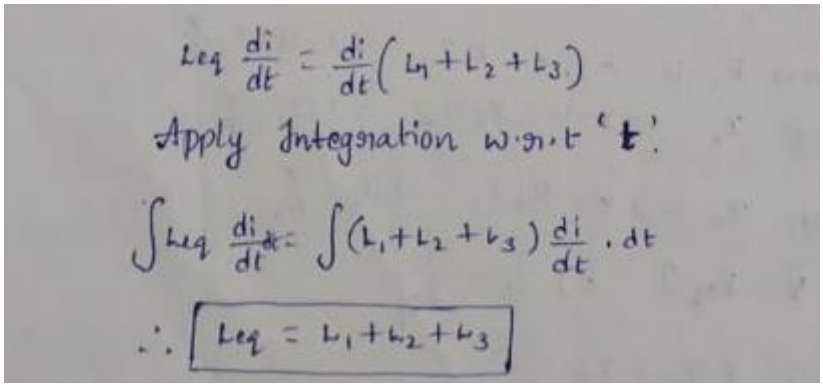
$Req = 5 + \frac{40}{7}$ → $\mathbf{R_{eq}} = \frac{75}{7}\ \Omega$

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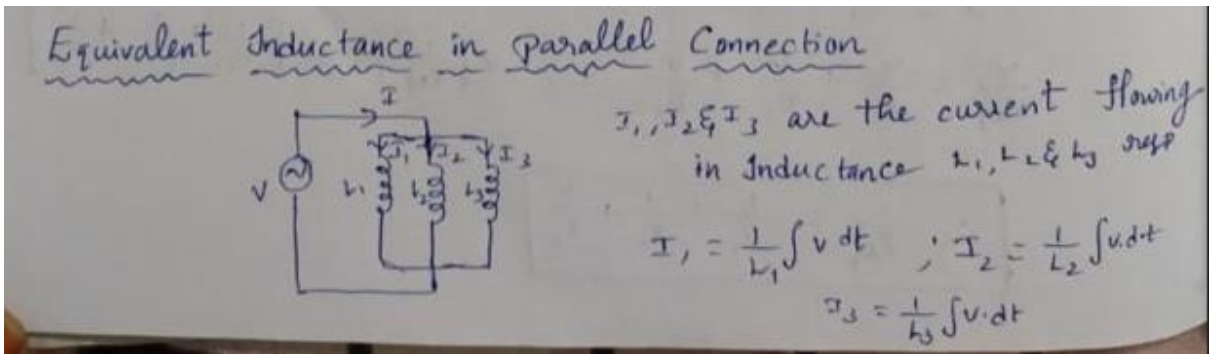
8. Inductors in Series

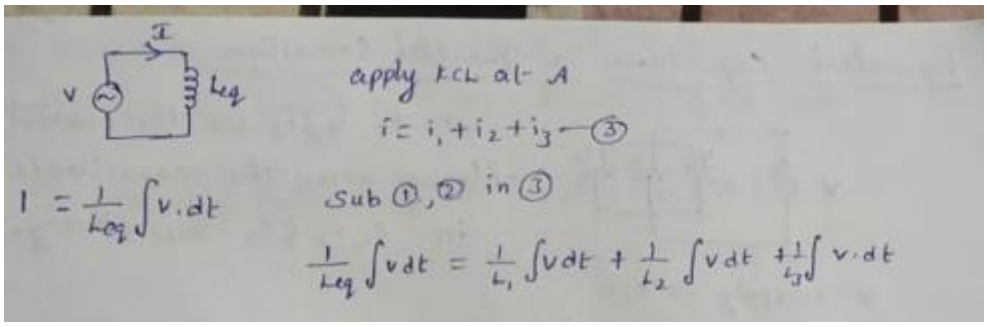


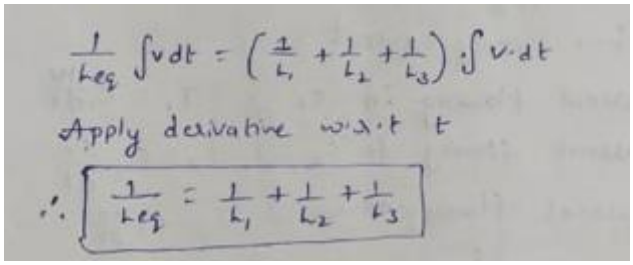




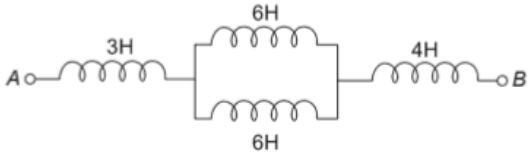
9. Inductance in parallel







10. Problem: Find the equvivalent L between A & B?

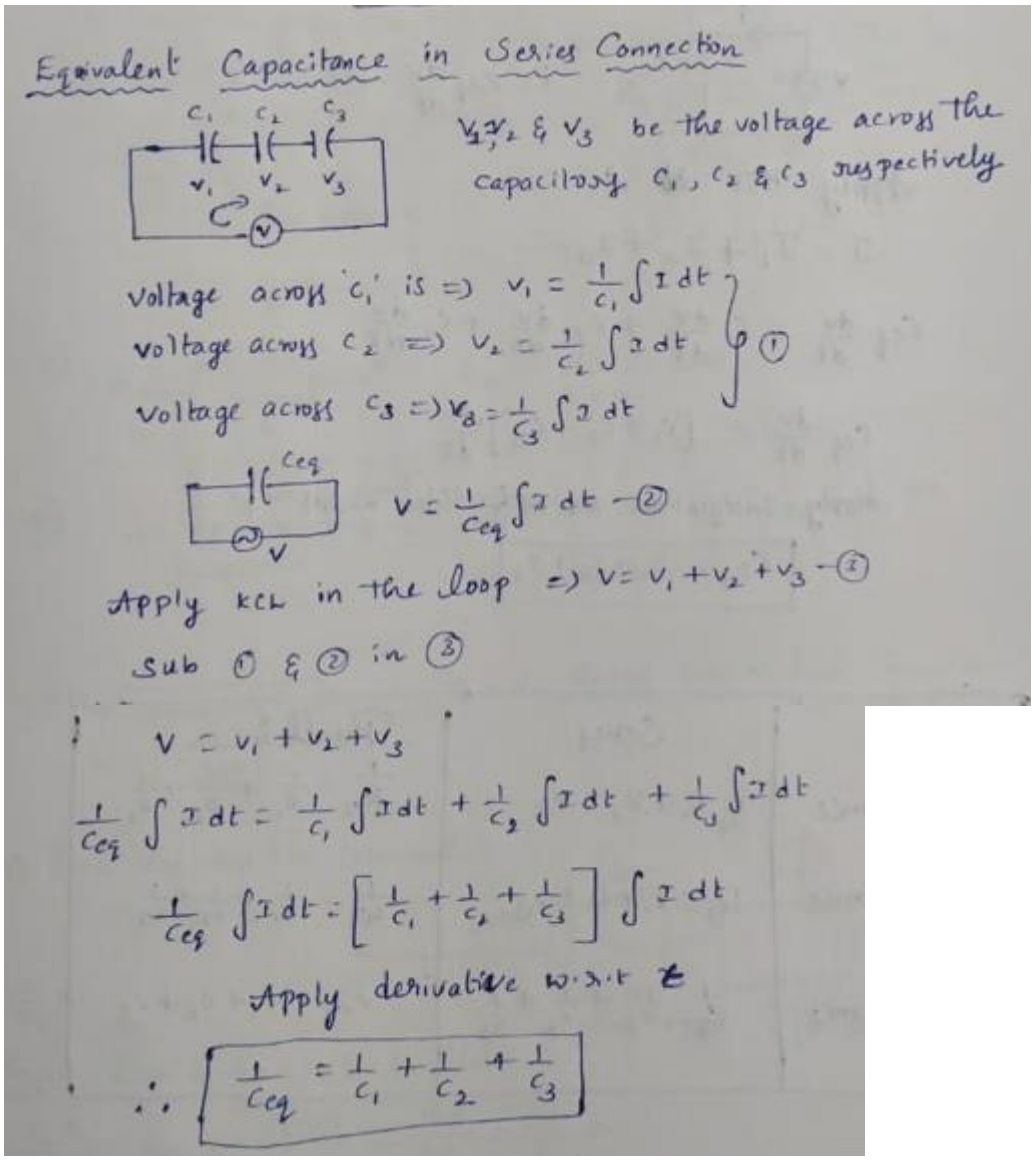


Solution:

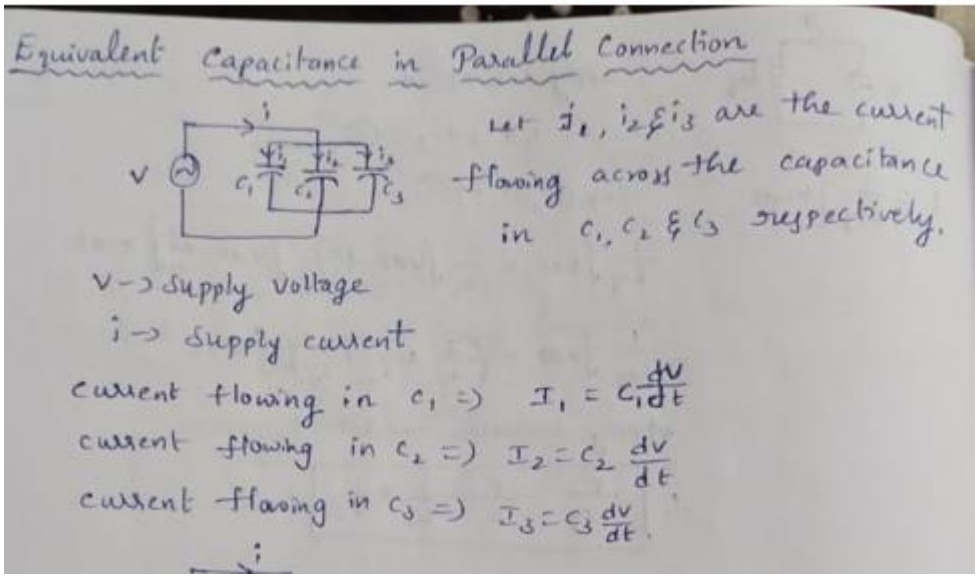
Step 1: 6 H & 6 H are in Parallel = $\frac{6\times 6}{6+6}$ = **3 H**

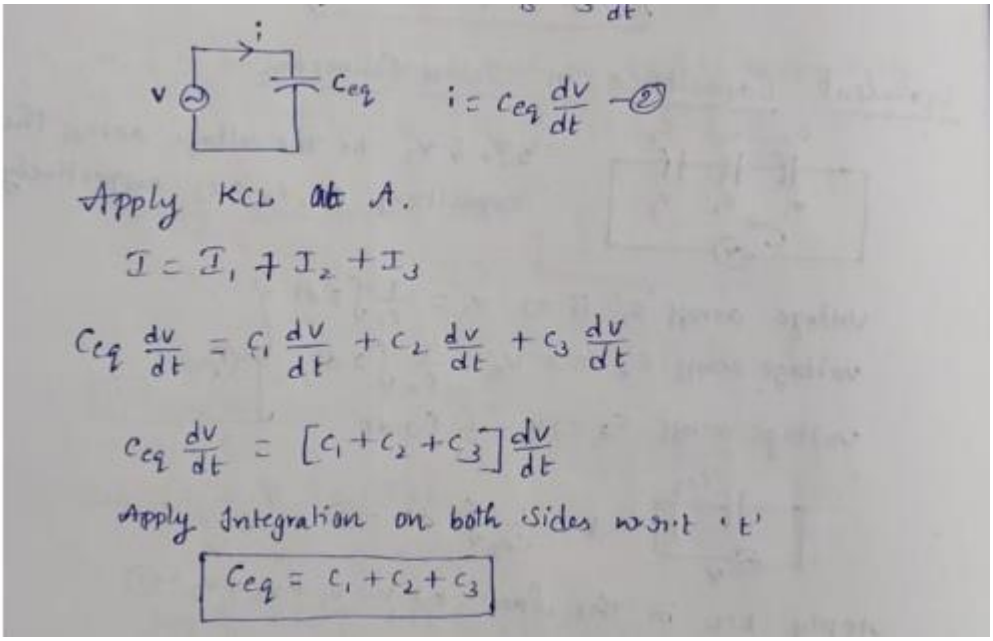
Step 2: 3 H , 3 H , 4 H are in series = 3 + 3 + 4 = 10 H

11. Capacitor in Series



12. Capacitors in Parallel

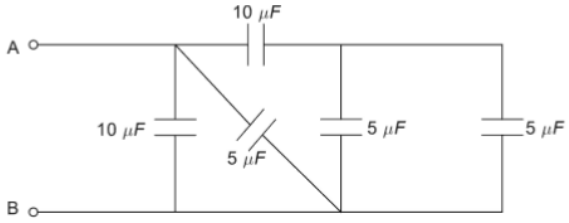




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13. Problem :

Find the equvivalent capacitance between A & B ?



solution:

Step 1: From the last end of circuit,

Capacitances 5 & 5 are in parallel = 5+5 = 10 μF

Step 2 : Then 10 & 10 become series $= \frac{10\times10}{10+10} = 5$ μF

Step 3 : Then 5 & 5 become parallel, = 5+5 = 10 μF

Step 4 : Then 10 & 10 become parallel = 10 + 10 = 20 μF

The answer is C_t= 20 μF

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14. For Remembrance:

Elements Symbol	RESISTOR	CAPACITOR	INDUCTOR
Denoted by	R	C	L
Equation	$R = \frac{V}{I}$	$C = \frac{Q}{V}$	$L = \frac{V_L}{(di/dt)}$
Series	$R_T = R_1 + R_2$	$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2}$	$L_T = L_1 + L_2$
Parallel	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$	$C_T = C_1 + C_2$	$\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2}$

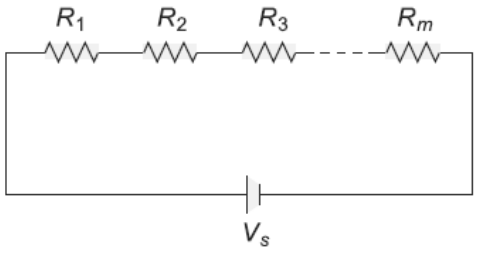
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15. Current division and Voltage division

i) Voltage Division Rule

- The series circuit acts as a voltage divider.
- Since the same current flows through each resistor, the voltage drops are proportional to the values of resistors.
- Using this principle, different voltages can be obtained from a single source, called a voltage divider
- If the circuit consists of a number of series resistors, the total current is given by the total voltage divided by equivalent resistance

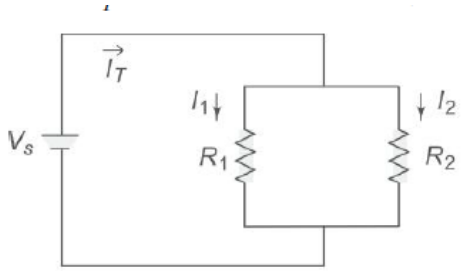


$$I = V_s/(R_1 + R_2 + \ldots + R_m)$$

ii) Current Division

- The total current entering into the parallel branches is divided into the branches currents according to the resistance values
- The branch having higher resistance allows lesser current, and the branch with lower resistance allows more current.
- The voltage applied across each resistor is Vs. The current passing through each resistor is given by

Current divider rule is used in parallel circuits to find the branch currents if the total current is known.



$$I_1 = \frac{V_s}{R_1},\ I_2 = \frac{V_s}{R_2}$$

16. Generation of sinusoidal voltage

Consider a coil of N turns as shown in the following figure (2.1). A pole of flux Φ Wb is initially at t = 0 is kept along the coil axis and then rotated with uniform angular velocity of ω rad/sec. At any instant 't' seconds later, pole moves through the angle $\theta = \omega t$ or radians. Now, flux Φ passes through the coil at an angle $\theta = \omega t$.

So, linkage flux = Flux along axis of coil = $\Phi \cos \omega t$.

For 'N' turns, Flux linkages = $N\Phi \cos \omega t$

According to Faraday's Law of electromagnetic induction instantaneous induced voltage at instant 't' is v(t) and is given by

$V(t) = \frac{-d}{dt}(flux\ linkages) = \frac{-d}{dt}(N\Phi \cos \omega t) = N\Phi \frac{-d}{dt}(\cos \omega t) = N\Phi\, \omega \sin \omega t$

$V(t) = N\Phi\, \omega \sin \omega t$ ..... Equation -1

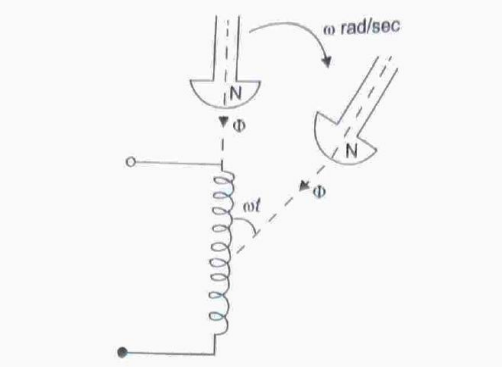


Figure (2.1): *Generation of emf in a rotating coil*

θ	0°	90°	180°	270°	360°
V(t)	0	$N\emptyset\omega$	0	$-N\emptyset\omega$	0

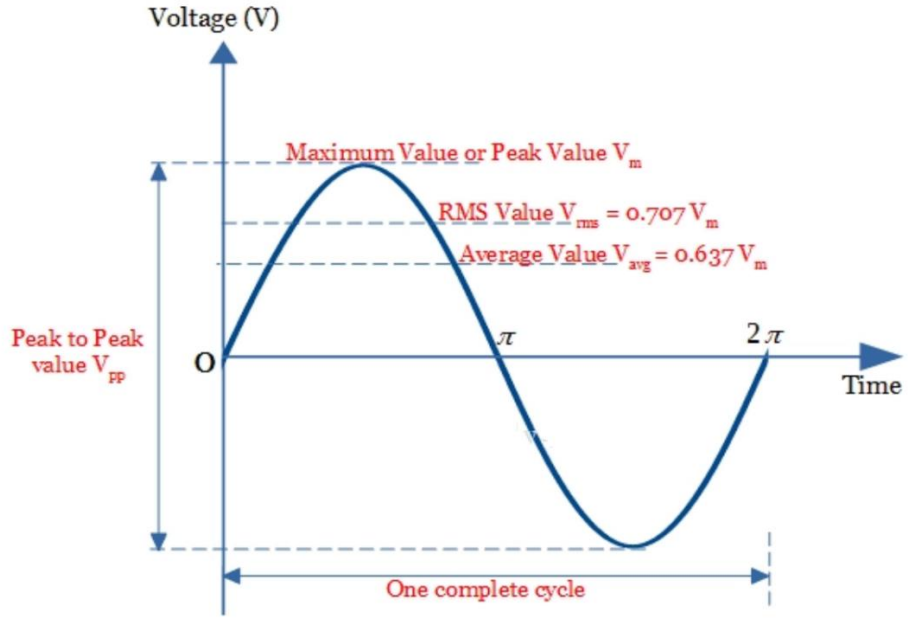
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17. Explain the following terms

a. Cycle b. Amplitude c. Frequency. d. Time period. e. Instantaneous value. f. Peak to peak value

a. Cycle.

One complete set of positive and negative values of an alternating quantity is known as a *cycle*. A cycle may also sometimes be specified in terms of angular measure. One complete cycle is said to spread over 360° or 2π radians.



b. Amplitude. The maximum value, positive or negative of an alternating quantity, is known as its *amplitude*.

c. Frequency (f). The number of cycles per second is called the frequency of the alternating quantity. Its unit is *hertz* (Hz).

d. Time Period (T). The time taken by an alternating quantity to complete one cycle is called its *time period*. It is measured in seconds.

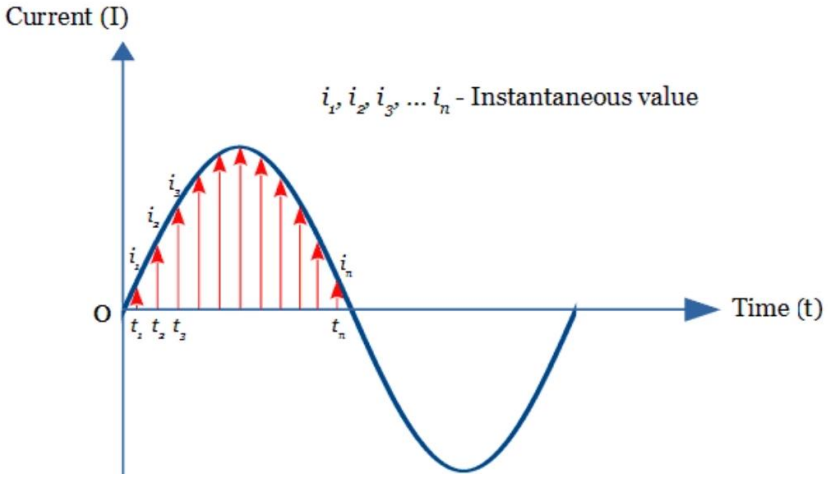
e. Instantaneous value.

The instantaneous value can be defined as the value of an alternating quantity at any particular instant. and is denoted for current as "i" and voltage as "e".

The instantaneous value of a sinusoidal waveform is generalized by the formula,

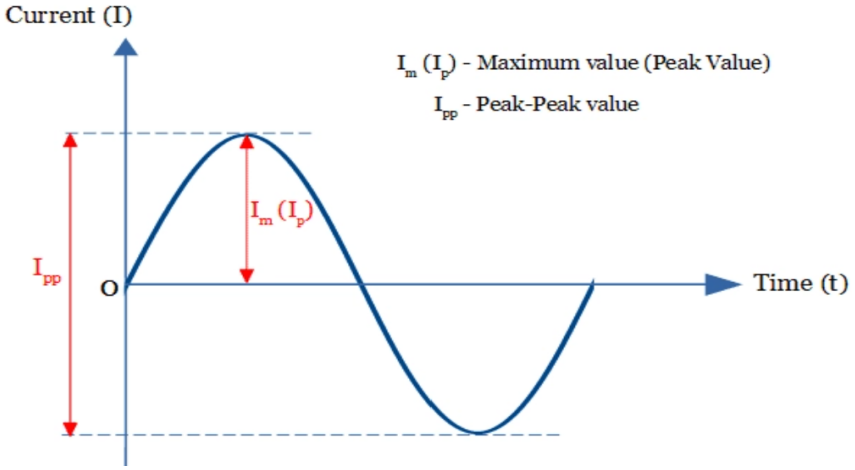
$$e = E_m sin\theta$$

$$i = I_m sin\theta$$

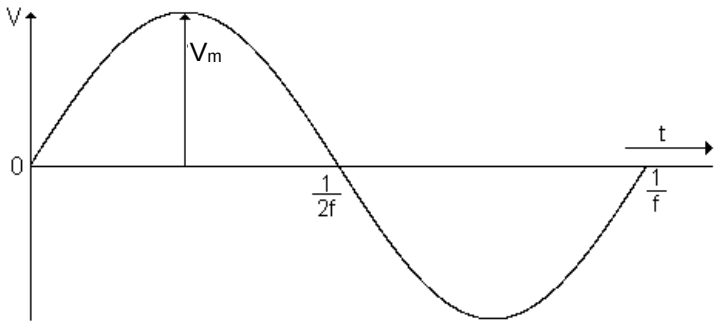


f. Peak to peak value

The sum of peak values of the positive and negative half cycle is known as peak-to-peak value. It is expressed as I_{pp}.



18. Derive the equation for Average value, RMS value, peak factor and form factor for sinusoidal waveform.



Where

Vm – peak applied voltage in volts

F= supply frequency in HZ

T= time in seconds

a. Average value

- The average of all instantaneous values during one alternation".
- Whereas the average value for the entire cycle of alternating quantity is zero, because of this average value for sinusoidal wave is calculated for half wave only.
- The instantaneous value at $t=1$ is i_1
- At $t = n$ is, i_n

- The average value for one alternation (0 to π) is

$$i_{avg} = \frac{i_1 + i_2 + i_3 + \ldots\ldots + i_n}{n}$$

- The average value of any function v(t), with period T is given by

$$v_{av} = \frac{1}{T}\int_0^T v(t)dt$$

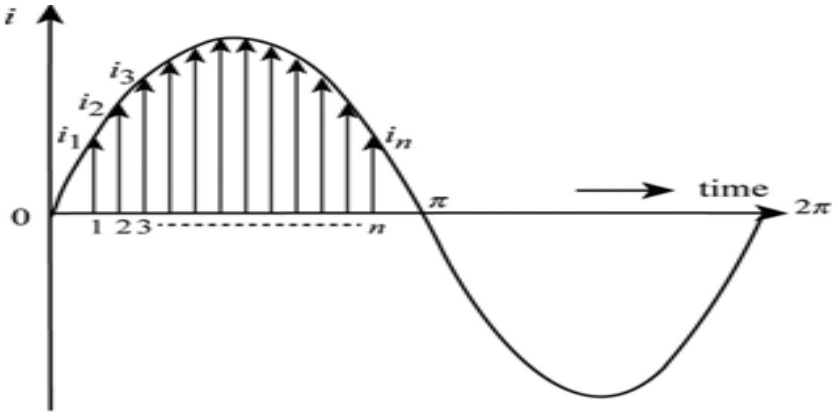
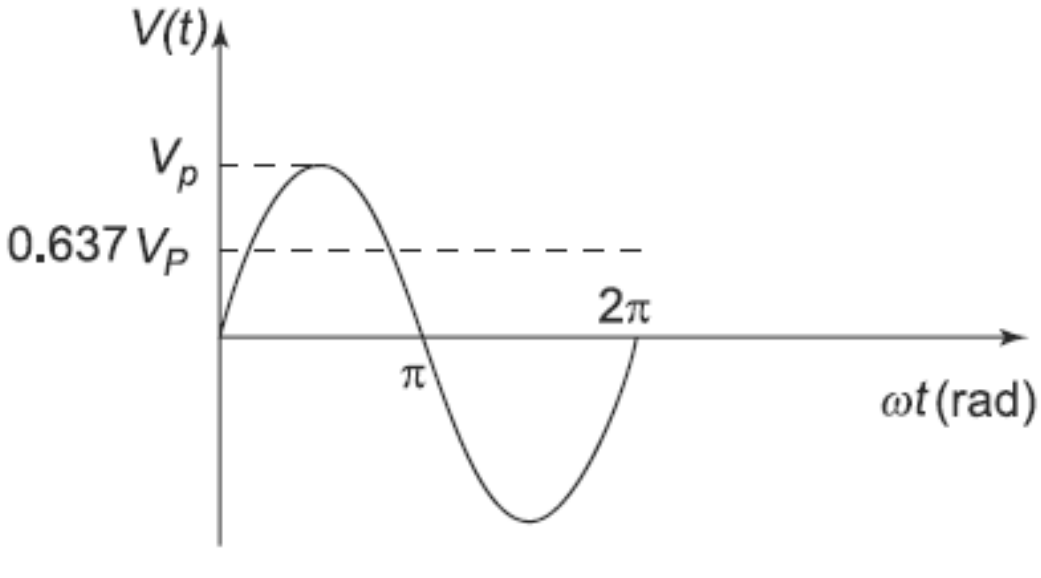


Figure 1

Derivation of average value for sin wave

The average value of the sine wave is the total area under the half-cycle curve divided by the distance of the curve.



Average value has to be calculated by considering one-half cycle as,

$$I_{av} = \frac{1}{\pi}\int_0^{\pi} I_m \sin\theta \, d\theta$$

$$= \frac{I_m}{\pi}\int_0^{\pi} I_m \sin\theta \, d\theta$$

$$= \frac{I_m}{\pi}[-\cos\theta]_0^{\pi}$$

$$= -\frac{I_m}{\pi}[-1-1]$$

$$= \frac{2\,I_m}{\pi} = 0.637\,I_m$$

Similarly average value for sinusoidal voltage, $V_{av} = 0.637\,V_m$

Therefore, average value $\frac{2\,I_m}{\pi}$ or $= \frac{2\,V_m}{\pi}$

b. RMS Value:

Definition

It is defined as root of the mean of the square of the periodic signal.
That means the rms value of a sine wave is equal to the dc voltage that produces the same heating effect.

$$I_{eff} = \sqrt{\frac{i_1^2 + i_2^2 + i_3^2 + \ldots\ldots + i_n^2}{n}}$$

$$I_{eff} = \sqrt{\text{mean of squares of instantaneous values}}$$

I_{eff} = square root of mean of squares of instantaneous values = R.M.S value

Derivation:

RMS current is defined as the square root of the mean of the squares of the instantaneous current over a cycle.

$$i = I_m \text{ Sin } \theta$$

Square of the current $i = I_m \sin\theta$ is $I_m^2 \sin^2\theta$

Its mean over one cycle is calculated by integrating it from 0 to 2π and dividing by the time period of 2π.

RMS value, $$I = \sqrt{1/2\pi \int_0^{2\pi} I_m^2 \sin^2\theta \, d\theta}$$

To calculate, let us put $$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$

then $$I = \sqrt{\frac{I_m^2}{4\pi} \int_0^{2\pi} (1-\cos 2\theta)\, d\theta}$$

$$= \sqrt{\frac{I_m^2}{4\pi}\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{2\pi}}$$

$$= \sqrt{\frac{I_m^2}{4\pi} \times 2\pi}$$

$$= \sqrt{\frac{I_m^2}{2}} = \frac{I_m}{\sqrt{2}}$$

Therefore, $$I = \frac{I_m}{\sqrt{2}} = 0.707\, I_m$$

$$\text{RMS value} = \frac{\text{Maximum value}}{\sqrt{2}} \quad \text{(for a sinusoidal wave)}$$

c. Form factor

It is the ratio of the RMS value to the average value of an alternating quantity.

$$K_f = \frac{\text{RMS value}}{\text{Average value}} = \frac{0.707\, I_m}{0.637\, I_m} = 1.11$$

d. Peak Factor:

The peak or crest factor is the ratio of peak or maximum value to its rms value

$$\text{Peak factor} = \frac{\text{Peak value}}{\text{RMS value}}$$

$$\text{Peak factor } (K_f) = \frac{I_m}{I_m/\sqrt{2}} = \sqrt{2} = 1.414.$$

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19. Explain the different types of power and power factor

There are three types of power are there, they are real power(P), reactive power (Q) and apparent power (S).

1. **Active power (P)**

- **Definition:** The power actually consumed by the circuit to perform work, such as heating a resistor or turning a motor's shaft. It is also known as real power or true power.
- **Usage:** It is the power that results in a net energy transfer that can be used.
- **Measurement:** Watts (W) or kilowatts (kW).

Active power, $P = VI \cos\emptyset$ Watts

2. Reactive power (Q)

- **Definition:** The power that flows back and forth between the source and load due to the presence of inductors and capacitors, but is not consumed. It is necessary for creating and maintaining magnetic and electric fields.
- **Usage:** It performs no net work but is crucial for the operation of certain devices like motors.
- **Measurement:** Volt-amperes reactive (VAR).

$Q = VI \sin\emptyset$ VAR

3. Apparent power (S)

- **Definition:** The total power flowing in a circuit, calculated as the product of the root mean square (RMS) voltage and current. It is the vector sum of active and reactive power.
- **Usage:** It represents the total power that the system must be designed to handle.

Measurement: Volt-amperes (VA).

Apparent power, S = Voltage × Current

$S = VI$ VA

Power factor:

It is ratio of real or active power (P) to the apparent power (S).

$$\text{Power Factor} = \cos\phi = \frac{\text{Real Power (P)}}{\text{Apparent Power (S)}}$$

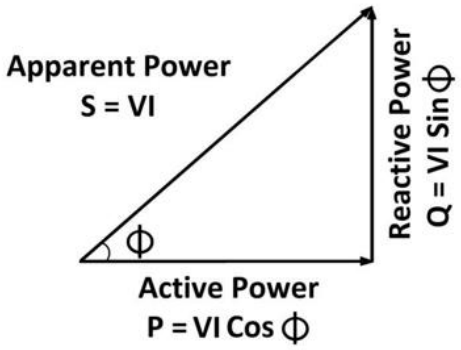
It is also defined as the cosine of the phase angle between the voltage and current. cosØ is known as power factor.

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20. Explain the power triangle

Power Triangle:

These three powers are shown in the power triangle of Figure.



Relationship between the three powers

- **Power Triangle:** The relationship between active, reactive, and apparent power can be visualized using the Pythagorean theorem in the form of a right-angled triangle.
- **Active Power : is the adjacent side.**
- **Reactive Power : is the opposite side**
- **Apparent Power : is the hypotenuse**

  As per the right triangle formula,

$$S^2 = P^2 + Q^2$$

$$S = \sqrt{P^2 + Q^2}$$

THE END